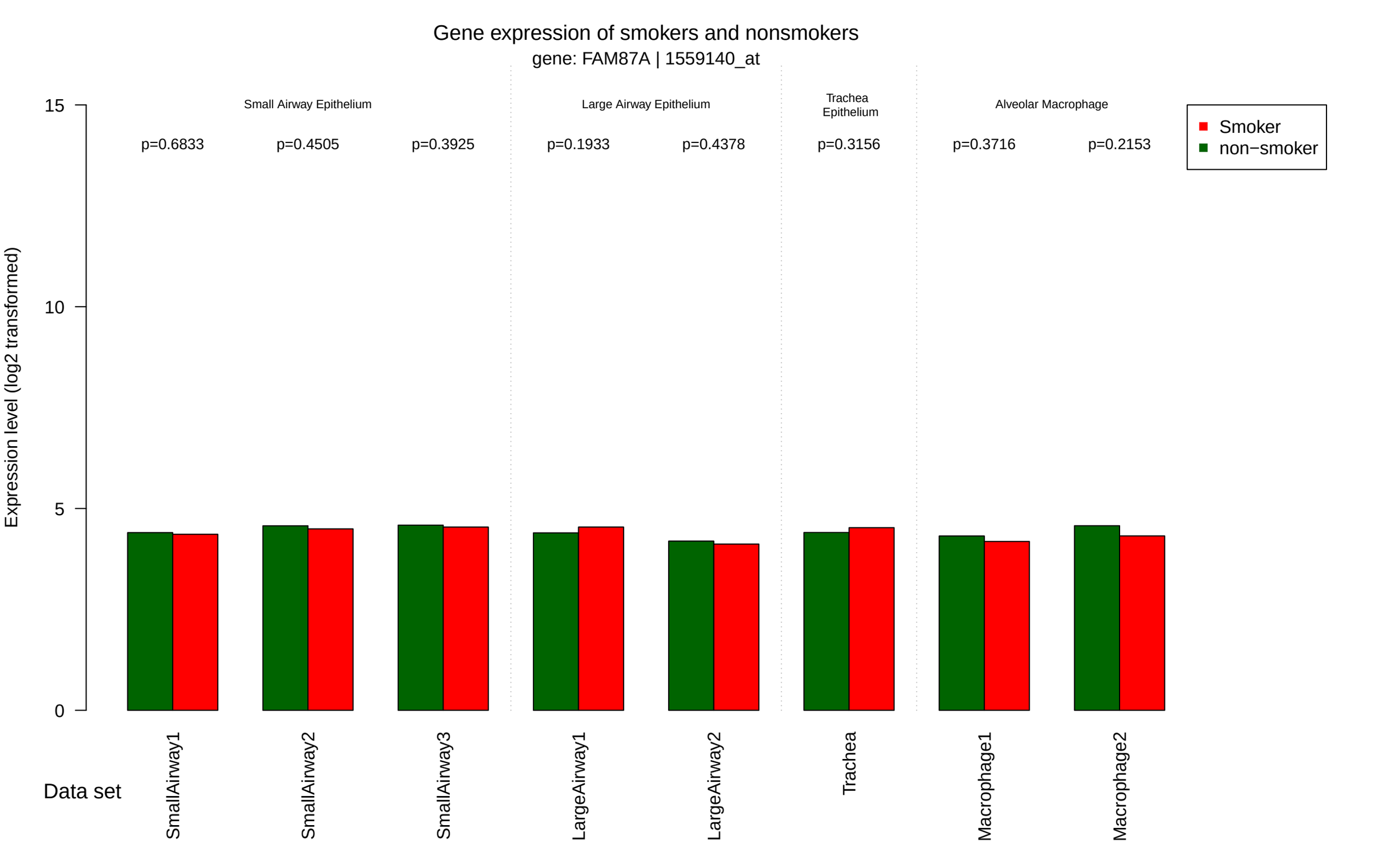
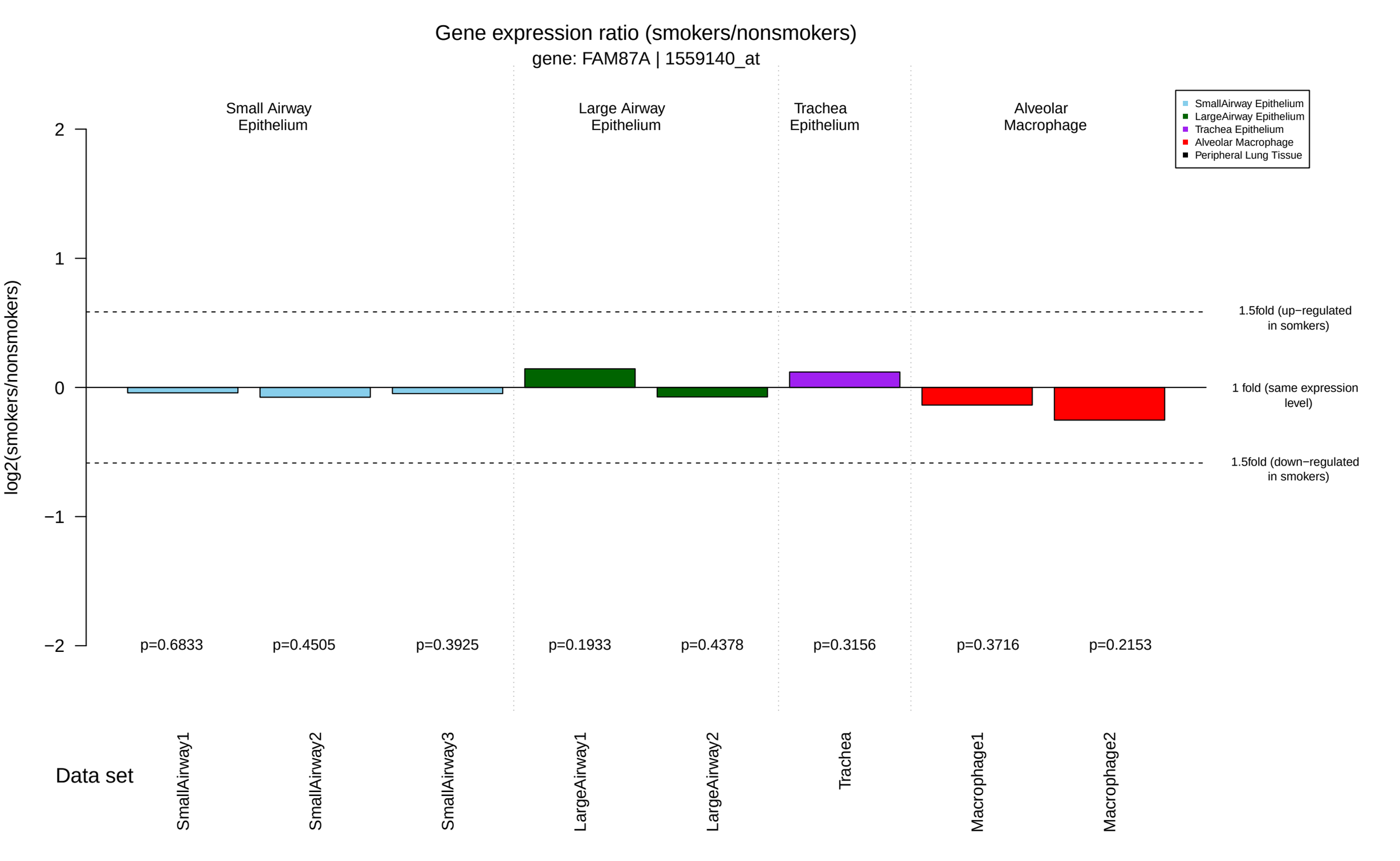
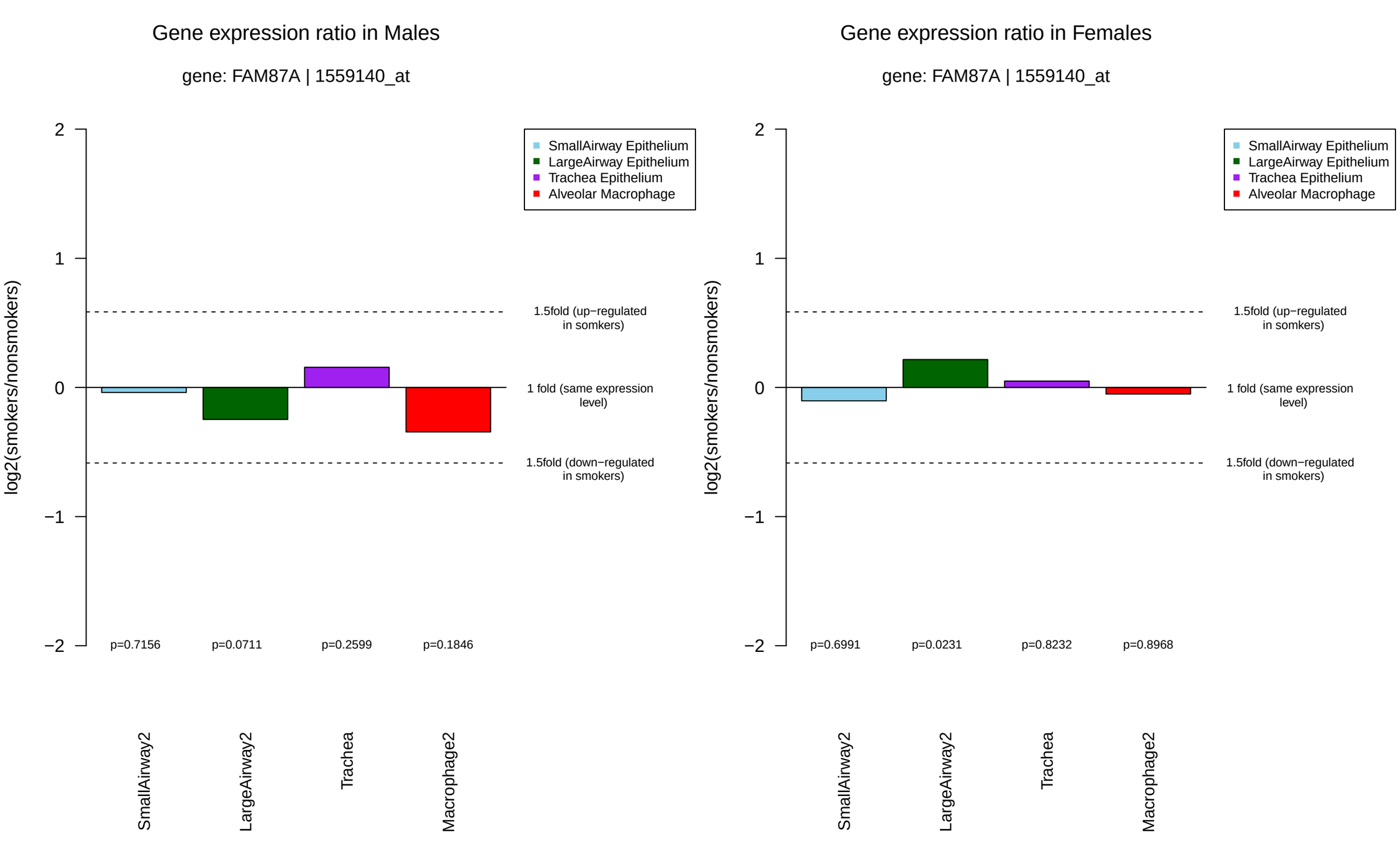
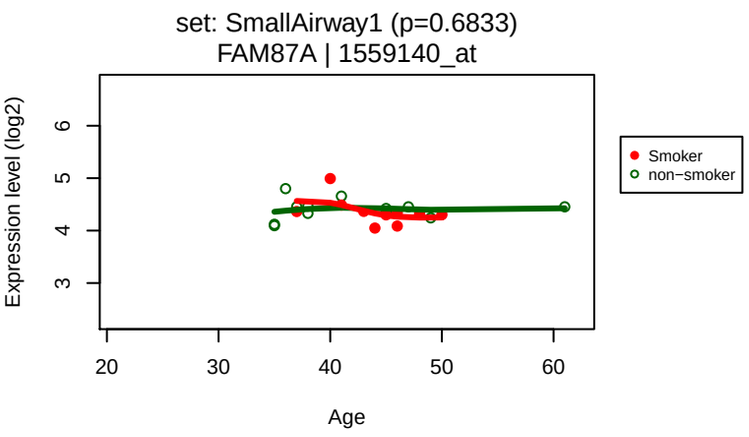
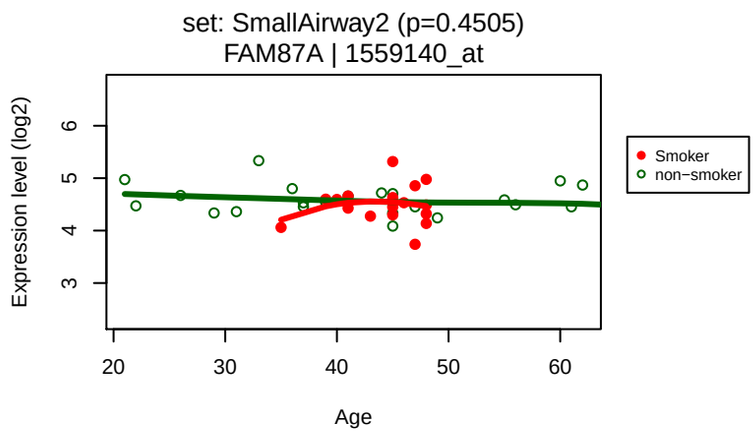
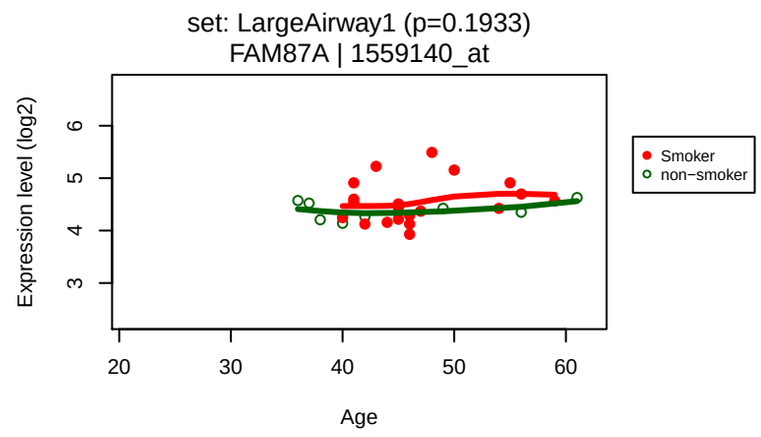

Gene expression of smokers and nonsmokers
gene: FAM87A | 1559140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6833
p=0.4505
p=0.3925
p=0.1933
p=0.4378
p=0.3156
p=0.3716
p=0.2153
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

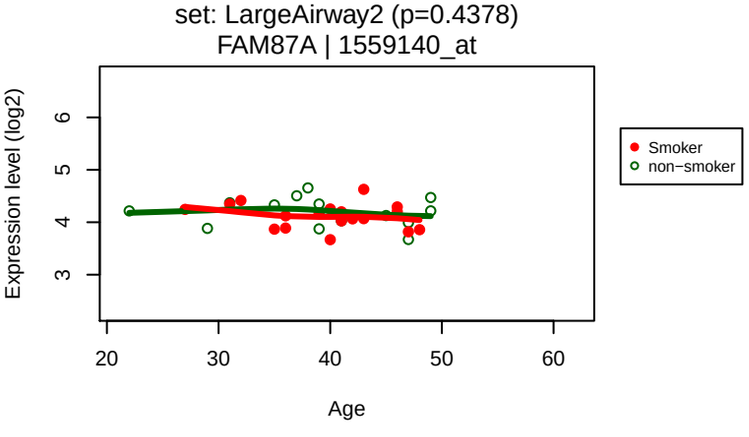

Gene expression ratio (smokers/nonsmokers)
gene: FAM87A | 1559140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6833
p=0.4505
p=0.3925
p=0.1933
p=0.4378
p=0.3156
p=0.3716
p=0.2153
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

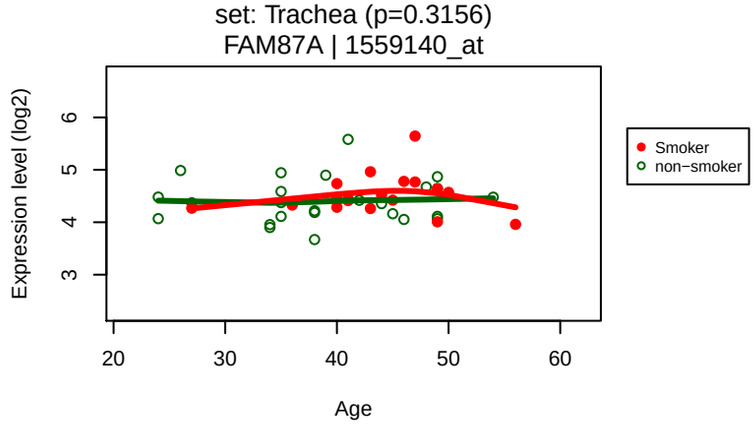

Gene expression ratio in Males
gene: FAM87A | 1559140_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7156
p=0.0711
p=0.2599
p=0.1846
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM87A | 1559140_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6991
p=0.0231
p=0.8232
p=0.8968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

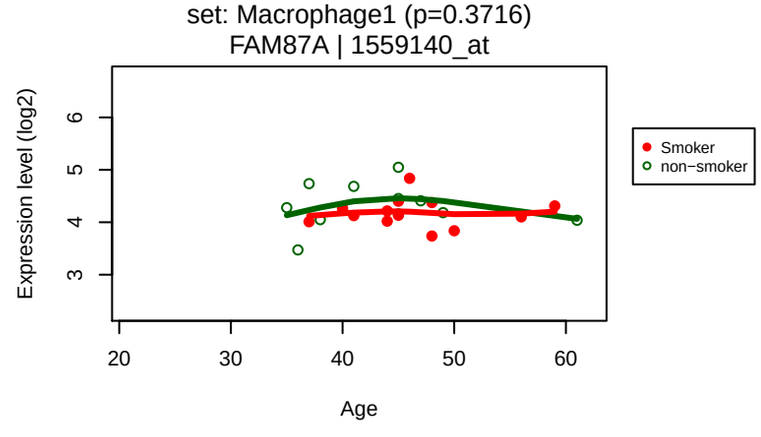

set: SmallAirway1 (p=0.6833)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker

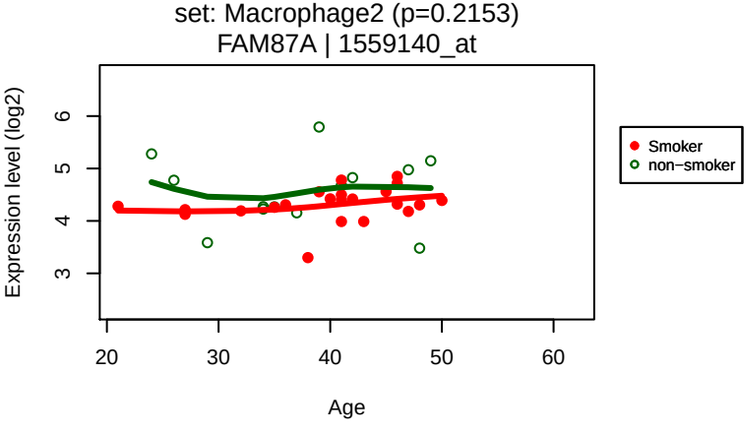

set: SmallAirway2 (p=0.4505)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker

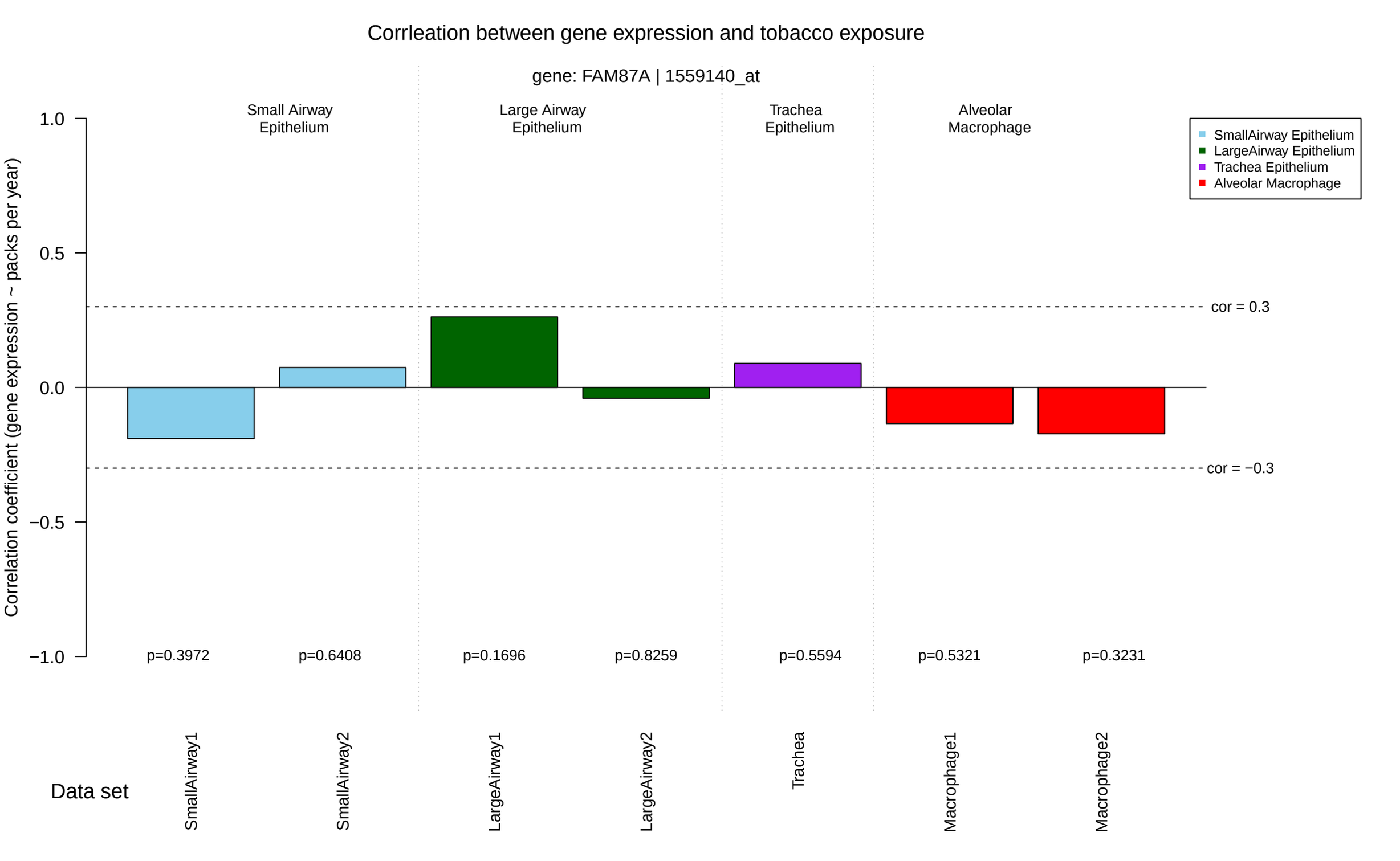

set: LargeAirway1 (p=0.1933)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4378)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3156)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3716)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2153)
FAM87A | 1559140_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM87A | 1559140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3972
p=0.6408
p=0.1696
p=0.8259
p=0.5594
p=0.5321
p=0.3231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2